Factor Dense-Boundary Convergence and Phase-4 Checkpoint

Tactix Development Project

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Executive summary

The current checkpoint closes the startup semantic mismatch in Tactix's Factor pipeline. Under the corrected dense-entry semantics, Tactix and Kissat select the same ordered Factor grids through all 766 startup grids observed on `mod4block_2vars_10gates_u2_autoenc-sc2009`. The implementation now preserves exact large-clause occurrences, assigned remainders, first-import heap order, the fixed same-pass factor boundary, and one unified local watch order. It also performs Kissat's first-watch-preserving duplicate-binary flush at the dense boundary.

This is a semantic checkpoint, not a performance closure. The correct trace-off Tactix arm times out at 60 seconds, while Kissat solves in 18.30 seconds. A superficially better Tactix arm solved in 45.89 seconds, but an adversarial fixture proved that it repeatedly counted the same binary candidate key. That arm is rejected. Direct-scan and cache variants also timed out and were removed under the project's arm-frugality rule.

The next causal frontier is now narrower. At the second Factor dose the solvers enter from already different formula and search states, at conflicts 2512 and 2513 respectively, and produce 16 versus 12 grids. Because startup Factor is exact, this second-dose difference is a downstream symptom. The next pass must compare the preceding search events one boundary at a time; Factor should be reopened only if equal input states expose an unequal grid, source reference, heap update, or effort stop.

1 Complete semantic unit implemented

Earlier trace alignment had hidden several coupled approximations. They were repaired together because each changes the mutable quotient consumed by the next Factor step.

1. Equal large clauses are no longer collapsed into content identity. Content is an equality key; each aligned row retains its exact clause reference, including repeated equal occurrences.
2. Root assignments no longer discard otherwise live remainders. Assigned literals may remain in a remainder, while the prospective factor itself must still be active and unassigned.
3. Tied Factor candidates are seeded in first-import literal order, reproducing Kissat's internal permutation without sacrificing Tactix's stable public identities.
4. Same-pass extensions may occur in remainders but may not become a new factor. The candidate boundary is fixed at phase entry.
5. Binary and large encounters use one ordered local stream: surviving binary partners, arena-order large occurrences, then same-pass additions. Deletions invalidate exact stream entries.

6. Large rows match the first unused exact source reference with the required remainder, rather than an arbitrary clause having equal content.

The final binary rule came from re-reading executable Kissat call sites. Dense entry flushes duplicate binary watches before Factor observes them, preserving the first watch for each literal–partner pair. Tactix formerly deduplicated only Factor's candidate view, leaving the physical watch state different for later rules and search. The new kernel operation performs that boundary transition once, keeps proof clauses live, and leaves the latest binary proof identifier available for naming. Focused tests pin both the first-watch order and the absence of duplicate Factor candidates.

2 Measured evidence and rejected arms

Table 1 separates accepted semantic evidence from diagnostic arms. Times are wall-clock results from the named fixed allocation; "timeout" means the 60-second limit was reached. SAT results cited as verified passed independent model checking.

Purpose	Artifact / job	Result and disposition
Unified startup order	unified-watch trace v2	711 exact grids in the original common window; semantic arm retained
Rejected overcount arm	job `1607031`	45.89 s versus Kissat 18.30 s; 864 grids and 126,212,303 Factor ticks, but adversarial 2-by-3 duplication proves candidate overcount; rejected
Correct dense boundary	job `1607035`	Tactix timeout at 60 s; Kissat 18.30 s verified; current production semantics
Post-startup Tactix trace	job `1607037`	766 exact startup grids; second dose at conflict 2512, 16 grids
Post-startup Kissat trace	job `1607038`	766 exact startup grids; second dose at conflict 2513, 12 grids
Direct/cache experiments	jobs `1607010`, `1607012`, `1607014`, `1607016`, `1607018`, `1607026`, `1607039`	Longer or equal trace prefixes but production timeout; all code removed

Table 1: Factor convergence evidence. Only the dense-boundary semantic arm is retained.

The 766-grid startup identity is measured. The statement that the second-dose grid difference is downstream is an inferred causal classification supported by the unequal conflict and formula/search entry states. It does not yet identify the earlier event that created those states. Likewise, the timeout demonstrates a cumulative performance regression but does not by itself attribute that regression to Factor's inner-loop cost.

3 Other retained work in this checkpoint

This checkpoint also carries the coupled Sweep/SubSolver and policy tracing work needed by the broader one-instance-at-a-time Phase-4 method. Sweep now uses the reference Kitten tick currency throughout its queries, flips, round checks, scheduler meter, and public effort counter. Its environment and model operations are represented as complete stateful units rather than local stand-ins. Substitution follow-ups and solve-entry root lifecycle behavior are pinned by focused tests. Default-strategy constants introduced by these mechanisms live in the configuration surface rather than as policy magic numbers.

Diagnostic event tracing remains an implementation feature only in explicit trace builds. The production configuration compiles out calls, event strings, and implementation symbols. The new traces

provide exact Factor grids, ordered members, source references, heap transitions, phase boundaries, and the adjacent solver state needed to decide whether a future mismatch belongs to Factor or to an earlier mechanism.

4 Verification at the checkpoint

The final source state passed the following local gates:

- the full ASan-enabled debug suite excluding the separately exercised scripted-default example: 68 of 68 tests;
- the release scripted-default example: 1 of 1;
- Factor, Factor-N1, kernel, solve-reset, proof micro-fixtures, configuration wiring, policy/rule boundary, documentation, and strategy configuration checks;
- the production zero-overhead trace test: 1 of 1;
- document line-budget and whitespace checks.

No current SAT experiment remains in Slurm. The allocation list contains two interactive shell sessions and an unrelated GPU job rooted in `/home/hwu/reduction`; none is modified by this wrap-up.

5 Next controlled increment

1. Start from the exact 766-grid startup checkpoint and identify the first unequal search/formula event before conflicts 2512/2513. Compare one event family or matched feature-off projection at a time.
2. Decide whether Tactix's behavior is demonstrably strictly better. If not, port the complete Kissat semantic unit; do not retain a proxy or another long-lived arm.
3. Require a reach counter and localized trace, focused proof/model fixtures, the earlier P4A fixtures as cumulative guards, a trace-off build, and a same-allocation production comparison.
4. Reopen Factor only if both solvers reach it with the same input state and nevertheless differ in grid order, exact source selection, update order, or budget termination.

The Phase-4 exit criterion is unchanged: Tactix must remain competitive with Kissat and CaDiCaL. The tightened methodology adds a second obligation: on each selected Tier-A/B fixture, the trajectory either matches Kissat or the first divergence has a localized, source-backed, technically justified reason.